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BSE-2A

Digital Logic Design (DLD), Assignment #1

1. Fuel Level Encoding and Calibration;

(i) Conversion; $(+45)_{10} = (00101101)_2$

2's Complement of $(45)_{10} = (11010011)_2$

Conversion factor $= (+12)_{10} = (00001100)_2$

(ii) Adding / Subtracting Correction factor;

$(-45 + 12)_{10} = (11010011 + 00001100)_2$

$$\begin{array}{r} 11010011 \\ + 00001100 \\ \hline 11011111 \end{array}$$

$= (11011111)_2$ (in 2's complement form)

(iii) Check for Overflow;

→ Since the resultant value lies between the range of the 2's complement form $(-128 \text{ to } 127)$ and the addition in binary had no complications, there is no overflow condition.

(iv) Convert back to Decimal;

$$\begin{aligned} (11011111)_2 &= (-2^7 + 2^6 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0)_2 \\ &= (-128 + 64 + 16 + 8 + 4 + 2 + 1)_2 \\ &= -33 \end{aligned}$$

2. 8-bit 2s Complement Addition and Overflow Handling;

(i) Conversion;

$$A = (11110101)_2 = (-2^7 + 2^4 + 2^5 + 2^4 + 2^2 + 2^0)_{10}$$

$$= (-128 + 64 + 32 + 16 + 4 + 1)_{10}$$

$$= (-11)_{10}$$

$$B = (11001100)_2 = (-2^7 + 2^6 + 2^3 + 2^2)_{10}$$

$$= (-128 + 64 + 8 + 4)_{10}$$

$$= (-52)_{10}$$

(ii) Binary Addition;

$$(11110101 + 11001100)_2$$

$$(A + B)$$

$$\begin{array}{r} 11110101 \\ + 11001100 \\ \hline 11000001 \end{array}$$

$$= (11000001)_2 \quad (\text{in 2s complement form})$$

(iii) Overflow Check;

$$\rightarrow (-11) + (-52) = -63 \quad (\text{and } 11000001 = -128 + 64 + 1 = -63)$$

→ The resultant $(11000001)_2$ lies between the range of 2s complement binary numbers, so no overflow occurs.

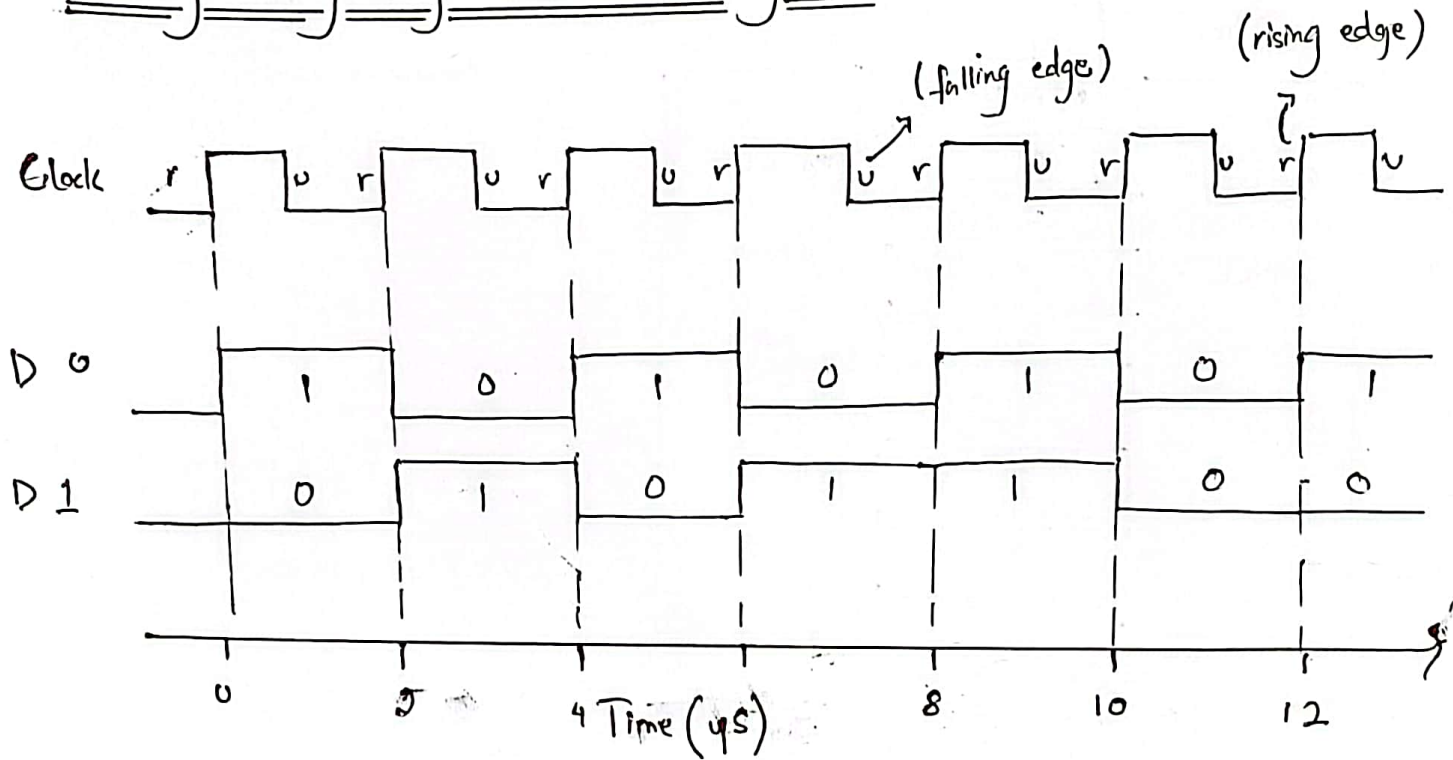
→ When performing addition the MSB stayed consistent which also indicates no overflow. (added 2 negative and answer was negative)

→ Storing this in a 16 bit register will simply extend the MSB.

$$= (1111111111000001)_{16}$$

3. Timing Diagram and Data Transfer;

(i) Drawing Timing Diagram with clock signal;



read : r

→ update

update : u

(ii) Explanation of Timing Diagram;

- The memory module places new data at each falling edge (u).
- The microcontroller reads the available data at each rising edge (r).
- This ensures stability and synchronized data transfer.
- The signals are synchronized with the bit time of the clock.

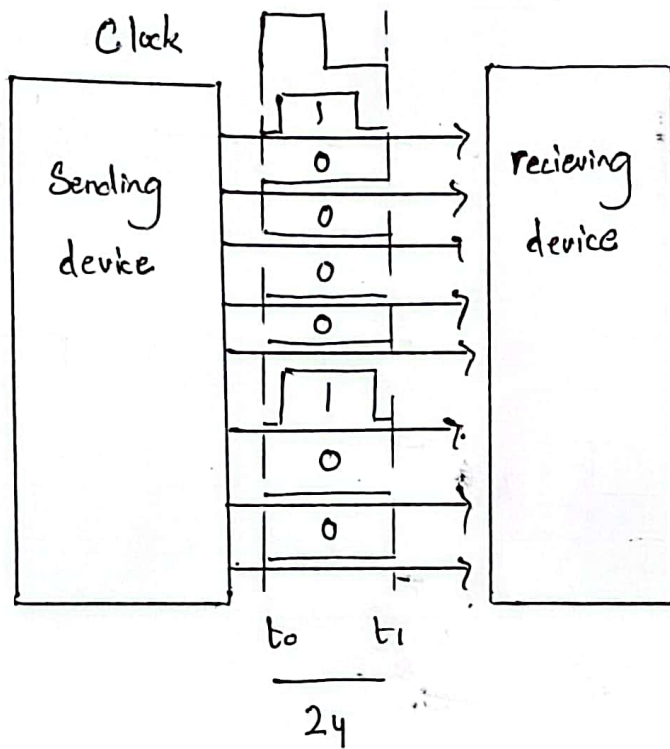
(iii) Questions;

(a) : Frequency of clock signal = $\frac{1}{T} = \frac{1}{2 \times 10^{-6}} = 500 \text{ kHz}$

(b) : The value must have been updated at the last falling edge.

(c) : microcontroller would read unstable data, partially updated or incorrect values. To ensure stability the memory should always update before the microcontroller reads.

(iv) Example of Parallel Transfer;



→ 8 bits transferred in one instance.